

5(a)	1 at 95 °C, (DNA) denatures / (dsDNA) splits into single strands / H bonds break ; 2 at 50 °C, primers, anneal / bind, (to ssDNA) ; 3 at 72 °C, complementary / second, (DNA) strand is made ; A DNA replicated / (ssDNA) is made into dsDNA	3
5(b)	any four from 1–7: 1 <i>M. pygmaea</i> (populations) geographically isolated / separated by sea or water ; 2 <i>M. pulvinaris</i> (populations) geographically isolated / mountain tops separated ; <i>M. pygmaea</i> or <i>M. pulvinaris</i> / <i>South Island</i> or <i>Stewart Island</i> species: 3 no / little, interbreeding / gene flow, between populations ; 4 different, selection pressures / environment ; 5 different / random, mutations ; 6 different changes in allele frequencies / different gene pools ; 7 (so) allopatric speciation could be occurring (within either species) / AW ; <i>M. pottsiana</i> / <i>North Island</i> species: 8 no geographical isolation ; 9 (so) interbreeding / gene flow, occurs, within / between, population(s) ; 10 similar, selection pressures / environment ;	5

Question	Answer	Marks
5(c)	any two from: 1 genetic drift / bottleneck / founder effect ; 2 migration ; 3 mutation ; 4 artificial selection / selective breeding ;	2